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Section: C

Date:

Assignment#3

Examples (6.1)

Example 1 laplace Transform

Let $f(t)=1$ when $t \geq 0$. Find $F(s)$

Solution:

From (1) we obtain by integration

$$L(f)=L(1)=\int_0^{\infty} e^{-st}\,dt=-\frac{1}{s}e^{-st}\Big|_0^{\infty}$$

$$=\frac{1}{s}$$

Such an integral is called an improper integral and, by definition, is evaluated according to the rule

$$\int_0^{\infty} e^{-st}f(t)\,dt=\lim_{T\to\infty}\int_0^{T} e^{-st}f(t)\,dt.$$

Hence our convenient notation means

$$\int_0^{\infty} e^{-st}\,dt=\lim_{T\to\infty}\left[-\frac{1}{s}e^{-st}\right]_0^{T}=$$

$$\lim_{T\to\infty}\left[-\frac{1}{s}e^{-sT}+\frac{1}{s}e^{0}\right]=\frac{1}{s} \quad (s>0)$$

we shall use this notation throughout this chapter

Example 2 laplace Transform $L(e^{at})$ of the Exponential Function e^{at}

Let $f(t)=e^{at}$ when $t \geq 0$, where a is a constant. Find $L(f)$

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solution

Again by (1),

$$L(e^{at}) = \int_0^\infty e^{-st} e^{at} dt = \frac{1}{a-s} e^{-(s-a)t}\Big|_0^\infty;$$

hence, when $s-a>0$,

$$L(e^{at}) = \frac{1}{s-a}$$

Examples (6.3)

Example 1

write the following function using unit step functions and find its transform.

$$f(t) = \begin{cases} 2 & \text{if } 0<t<1 \\ \frac{1}{2}t^2 & \text{if } 1<t<1/2\pi \\ \cos t & \text{if } t>1/2\pi \end{cases}$$

solution. step 1: In terms of unit step functions

$$f(t) = 2(1-u(t-1)) + 1/2t^2 (u(t-1) - u(t-1/2\pi)) + (\cos t)\, u(t-1/2\pi).$$

Indeed, $2(1-u(t-1))$ gives $f(t)$ for $0<t<1$, and so on.

step2.

To apply Theorem 1, we must write each term in $f(t)$ in the form $f(t-a)\,u(t-a)$. Thus, $2(1-u(t-1))$ remains as it is and gives the transform $2(1-e^{-s})/s$. Then

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$$L\left\{\frac{1}{2}t^2 u(t-1)\right\} = L\left(\frac{1}{2}(t-1)^2 + (t-1) + \frac{1}{2}\right)$$

$$\left. u(t-1)\right\} = \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right)e^{-s}$$

$$L\left\{\frac{1}{2}t^2 u\left(t-\frac{1}{2}\pi\right)\right\} = L\left\{\frac{1}{2}\left(t-\frac{1}{2}\pi\right)^2 + \frac{\pi}{2}\left(t-\frac{1}{2}\pi\right) + \frac{\pi^2}{8}\right) u \left(t-\frac{1}{2}\pi\right)\right\}$$

$$= \left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s}\right)e^{-\pi s/2}$$

$$L\left\{(\cos t)u\left(t-\frac{1}{2}\pi\right)\right\} = L\left\{-\left(\sin\left(t-\frac{1}{2}\pi\right)\right)u\left(t-\frac{1}{2}\pi\right)\right\} = -\frac{1}{s^2+1}e^{-\pi s/2}$$

Together,

$$L(f) = \frac{2}{s} - \frac{2}{s}e^{-s} + \left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right)e^{-s}$$

$$-\left(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s}\right)e^{-\pi s/2} - \frac{1}{s^2+1}e^{-\pi s/2}$$

If the conversion of $f(t)$ to $f(t-a)$ is inconvenient, replace it by

(4**) $L\{f(t)u(t-a)\} = e^{-as} L\{f(t+a)\}$.

(4**) follows from (4) by writing $f(t-a) = g(t)$

hence $f(t) = g(t+a)$ and then again writing f for g. Thus

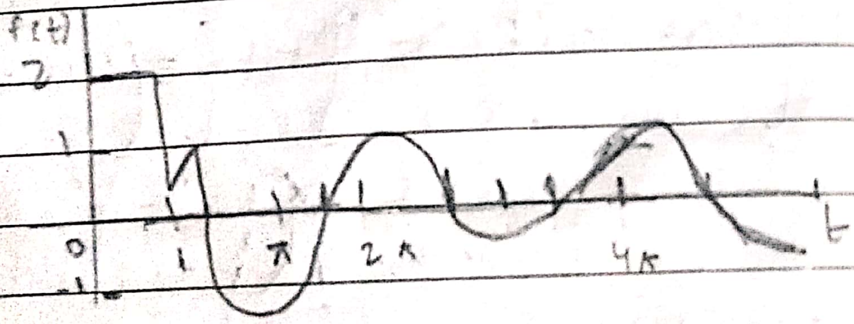
$$L\left\{\frac{1}{2}t^2 u(t-1)\right\} = e^{-s} L\left\{\frac{1}{2}(t+1)^2\right\} = e^{-s} L\left\{\frac{1}{2}t^2 + t + \frac{1}{2}\right\} = e^{-s}\left(\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\right)$$

as before, similarly for $L\{\frac{1}{2}t^2 u(t-\frac{1}{2}\pi)\}$.

Finally, by (4**),

$$L\left\{\cos t\, u\left(t-\frac{1}{2}\pi\right)\right\} = e^{-\pi s/2} L\left\{\cos\left(t+\frac{1}{2}\pi\right)\right\}$$

$$= e^{-\pi s/2}\frac{1}{s^2+1}$$



Example 2:

Find the inverse transform $f(t)$ of

$$F(s) = \frac{e^{-s}}{s^2+\pi^2} + \frac{e^{-2s}}{s^2+\pi^2} + \frac{e^{-3s}}{(s+2)^2}$$

Solution

without the exponential function in the numerator the three terms of $F(s)$ would

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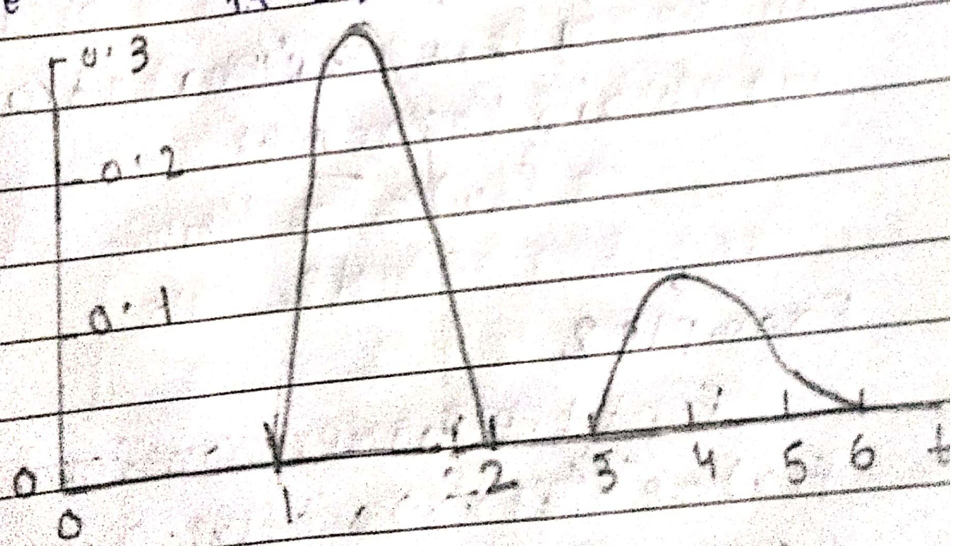
have the inverses $(\sin \pi t)/\pi$, $(\sin \pi t)/\pi$ and te^{-2t} because $1/s^2$ has the inverse t, so that $1/(s+2)^2$ has the inverse te^{-2t} by the first shifting theorem in sec 6.1. Hence by the second shifting theorem (t-shifting)

$$f(t) = \frac{1}{\pi}\sin(\pi(t-1))u(t-1) + \frac{1}{\pi}\sin(\pi(t-2))u(t-2) + (t-3)e^{-2(t-3)}u(t-3).$$

Now $\sin(\pi t - \pi) = -\sin \pi t$ and $\sin(\pi t - 2\pi) = \sin \pi t$, so that the first and second terms cancel each other when $t > 2$. Hence we obtain $f(t) = 0$ if $0 < t < 1$, $-(\sin \pi t)/\pi$ if $1 < t < 2$, 0 if $2 < t < 3$, and $(t-3)e^{-2(t-3)}$ if $t > 3$. See fig 123



Fg 123 f(t) in Example 2

Examples (11.9)

Example 1 Fourier Transform

Find the Fourier transform of $f(x) = 1$ if $|x| < 1$ and $f(x) = 0$ otherwise.

Solution

Using (6) and integrating, we obtain

$$\hat{f}(w) = \frac{1}{2\pi}\int_{-1}^{1} e^{-iwx}\,dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-iwx}}{-iw}\Bigg|_{-1}^{1}$$

$$= \frac{1}{iw\sqrt{2\pi}}(e^{-iw} - e^{iw})$$

As in (3), we have $e^{iw} = \cos w + i\sin w$, $e^{-iw} = \cos w - i\sin w$, and by subtraction

$$e^{iw} - e^{-iw} = 2i\sin w.$$

Substituting this in the previous formula on the right, we see that i drops out and we obtain the answer

$$\hat{f}(w) = \sqrt{\frac{\pi}{2}}\,\frac{\sin w}{w}$$

Example 2

Find the Fourier transform $\mathcal{F}(e^{-ax})$ of $f(x) = e^{-ax}$ if $x > 0$ and $f(x) = 0$ if $x < 0$; here $a > 0$

Solution. From the definition (6) we obtain by integration

$$\mathcal{F}(e^{-ax}) = \frac{1}{2\pi}\int_{0}^{\infty} e^{-ax}\, e^{-iwx}\,dx$$

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$$= \frac{1}{\sqrt{2\pi}} \frac{e^{-(a+iw)x}}{-(a+iw)} \Big|_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}(a+iw)}$$

Fourier series

Q) $f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi \end{cases}$

Solution

let

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot \cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot \sin nx\,dx$$

For a_0

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} 0 + \int_{-\pi/2}^{\pi/2} dx + \int_{\pi/2}^{\pi} 0\,dx\right]$$

$$a_0 = \frac{1}{\pi}\left[0 + \int_{-\pi/2}^{\pi/2} dx + 0\right]$$

$$a_0 = \frac{1}{\pi}(x)_{-\pi/2}^{\pi/2}$$

$$a_0 = \frac{1}{\pi}\left[\frac{\pi}{2} - \left(\frac{-\pi}{2}\right)\right] = \frac{1}{\pi}\left[\frac{\pi}{2} + \frac{\pi}{2}\right] = \frac{\cancel{2\pi}}{\cancel{2\pi}}$$

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$a_0 = 1$

For an

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx \, dx$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} 0 \cdot \cos nx \, dx + \int_{-\pi/2}^{\pi/2} (1)\cos nx \, dx \right.$$

$$\left. + \int_{\pi/2}^{\pi} 0 \cdot \cos nx \, dx\right]$$

$$a_n = \frac{1}{\pi}\left[0 + \int_{-\pi/2}^{\pi/2} \cos nx \, dx + 0\right]$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi/2}^{\pi/2} \cos nx \, dx\right]$$

$$a_n = \frac{1}{\pi}\left(\frac{\sin nx}{n}\right)_{-\pi/2}^{\pi/2}$$

$$a_n = \frac{1}{n\pi}\left[\sin nx\right]_{-\pi/2}^{\pi/2}$$

$$a_n = \frac{1}{n\pi}\left[\sin\frac{n\pi}{2} - \sin\left(-\frac{n\pi}{2}\right)\right]$$

$$a_n = \frac{1}{n\pi}\left[\sin\frac{n\pi}{2} + \sin\frac{n\pi}{2}\right]$$

$$a_n = \frac{1}{n\pi} 2\sin\frac{n\pi}{2}$$

$$a_n = \frac{2}{n\pi}$$

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For bn

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$b_n = \frac{1}{\pi}\left[\int_{\pi}^{-\pi/2} 0\cdot\sin nx\,dx + \int_{-\pi/2}^{\pi/2} 1\cdot\sin nx\,dx\right.$$

$$\left.+\int_{\pi/2}^{\pi} 0\cdot\sin nx\,dx\right]$$

$$b_n = \frac{1}{\pi}\left[0+\int_{-\pi/2}^{\pi/2}\sin nx\,dx + 0\right]$$

$$b_n = \frac{1}{\pi}\left(\int_{-\pi/2}^{\pi/2}\sin nx\,dx\right)$$

$$b_n = \frac{1}{\pi}\left(\frac{-\cos nx}{n}\right)_{-\pi/2}^{\pi/2}$$

$$b_n = \frac{1}{n\pi}\left(-\cos\frac{n\pi}{2} - \left(-\cos\frac{n\pi}{2}\right)\right)$$

$$b_n = \frac{-1}{n\pi}\left(\cancel{\cos\frac{n\pi}{2}} - \cancel{\cos\frac{n\pi}{2}}\right)$$

$$b_n = 0$$

$$f(x) = \frac{1}{2}1 + \sum_{n=1}^{\infty}\left[\frac{2}{n\pi}\cos nx + 0\sin nx\right]$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\left(\frac{2}{n\pi}\cos nx\right)$$

$$f(x) = \frac{1}{2} + \frac{2}{1\pi}\cos x + \frac{2}{2\pi}\cos 2x + \frac{2}{3\pi}\cos 3x + \frac{2}{4\pi}\cos 4x$$

$$+\frac{2}{5\pi}\cos 5x + \ldots\ldots$$

Q $f(x) = x \quad (0 < x < 2\pi)$

Solution

let

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

For a_0

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$a_0 = \frac{1}{\pi} \int_{2\pi}^{0} x\, dx$$

$$a_0 = \frac{1}{\pi} \int_{2\pi}^{0} \frac{x^2}{2}$$

$$a_0 = \frac{1}{\pi} \left. \frac{x^2}{2} \right|_{2\pi}^{0}$$

$$a_0 = \frac{1}{\pi} \left(\frac{0^2}{2} - \frac{(2\pi)^2}{2} \right)$$

$$a_0 = \frac{1}{\cancel{\pi}} \left(\frac{\cancel{4}\pi^{\cancel{2}}}{\cancel{2}} \right)$$

$$a_0 = 2\pi$$

For a_n

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos nx\, dx$$

$$a_n = \frac{1}{\pi} \int_{2\pi}^{0} x \cdot \cos nx\, dx$$

$$a_n = \frac{1}{\pi} \left[\int \left(x \cdot \frac{\sin nx}{n} \right) - \int (1) \frac{\sin nx}{n} \right]_{0}^{2\pi}$$

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$$a_n = \frac{1}{n\pi}\left[x \cdot \sin nx\right]_0^{2\pi} - \frac{1}{n\pi}\int \sin nx\,dx$$

$$a_n = \frac{1}{n\pi}\left[2\pi \sin 2\pi - 0\sin(0)\right] - \frac{1}{n\pi}\left[\frac{-\cos nx}{n}\right]_0^{2\pi}$$

$$a_n = \frac{1}{n^2\pi}\left[\cos(2\pi) - \cos n(0)\right]$$

$$a_n = \frac{1}{n^2\pi}(1-1) = 0$$

For bn

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cdot \sin nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{2\pi}^{0} x \cdot \sin nx\,dx$$

$$b_n = \frac{1}{\pi}\left[x\left(\frac{-\cos nx}{n}\right) - \int (1)\left(\frac{-\cos nx}{n}\right)dx\right]_0^{2\pi}$$

$$b_n = \frac{-1}{n\pi}\left[x\cos nx\right]_{2\pi}^{0} - \frac{1}{n\pi}\int_{2\pi}^{0} \cos nx\,dx$$

$$b_n = -\frac{1}{n\pi}\left[0 \cdot \cos 0 - 2\pi\cos(0)\right] - \frac{1}{n\pi}\left(\frac{\sin nx}{n}\right)_{2\pi}^{0}$$

$$b_n = -\frac{1}{n\pi}(0 + 2\pi(1)) + \frac{1}{n^2\pi}(\sin 0 - \sin 2\pi)$$

$$b_n = -\frac{1}{n\cancel{\pi}}(2\cancel{\pi})$$

$$b_n = -\frac{2}{n}$$

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$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$f(x) = \frac{1}{2} (2\pi) + \sum_{n=1}^{\infty} (0 \cdot \cos nx + b_n \sin nx)$$

$$f(x) = \pi + \sum_{n=1}^{\infty} (b_n \sin nx)$$

$$f(x) = \pi + (b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + b_4 \sin 4x + b_5 \sin 5x + \cdots)$$

$$f(x) = \pi - 2\sin x + (-1)\sin 2x - \frac{2}{3}\sin 3x - \frac{1}{2}\sin 4x - \frac{2}{5}\sin 5x + \cdots\cdots$$